80.3K ★

OPEN-SOURCE RAG · V0.25

WHAT IS RAGFLOW?

THE RAG ENGINE FOR REAL DOCS

Your parser is silently destroying answers before your model gets a chance. RAGFlow rebuilds the document layer with DeepDoc, OCR, table-structure recognition, and 11 chunking templates — then bolts on GraphRAG and agents.

- DEEPDOC · OCR · TSR
- 11 CHUNKING TEMPLATES
- GROUNDED VISUAL CITATIONS
- GRAPHRAG + AGENTS + MCP
- 80.3K ★ · 9.2K FORKS

PULKIT TYAGI
Senior Gen-AI Engineer

WHY NAIVE RAG BREAKS

~80% of enterprise data is unstructured. Generic **character-count splitters** hand the model garbage on the exact documents your business actually runs on.

- **Scanned PDFs** — OCR confidence collapses on rotated pages; tables become character soup before they reach the embedder
- **Tables** — spanning cells, merged headers, and projected row headers flatten into unreadable text strings
- **Multi-column layouts** — research papers, decks, and reports lose reading order; paragraphs interleave across columns
- **Citations** — auditors, lawyers, and analysts can't trace an answer back to a specific source region or page
- **Cross-doc reasoning** — naive top-k vector recall misses related-but-reworded facts spread across files

PULKIT TYAGI
Senior Gen-AI Engineer

MEET DEEPDOC

RAGFlow's in-house parser — the piece most frameworks outsource to **PyPDFLoader** and then quietly fail at. Three subsystems run in series on every page.

> *OCR pulls the text. Layout recognition tags 10 region types — Text, Title, Figure, Caption, Table, Header, Footer, Reference, Equation. TSR rebuilds tables cell-by-cell across rows, columns, and spanning cells.*
>
> — DeepDoc pipeline

Rotated scans? Confidence-scores OCR at 0°, 90°, 180°, 270° and re-extracts at the best angle. **Alt parsers?** MinerU and Docling are first-class options.

PULKIT TYAGI
Senior Gen-AI Engineer

TEMPLATE-BASED CHUNKING

Eleven parsers tuned to document *type*. **Pick the right one** per knowledge base — retrieval quality climbs before you tune anything else.

TEMPLATE	USE CASE
Laws	Preserve clause hierarchy & cross-refs
Paper	Keep abstract → method → results spine
Manual	Section-aware tech docs and SOPs
Book	Chapter & subsection scoped chunks
Presentation	Slide-level chunks with image context
Resume	Field-level extraction (skills, roles, dates)
Q&A	Pair-aware splitting for FAQs & tickets
Table / Picture	Structured rows; image + caption pairs

PULKIT TYAGI
Senior Gen-AI Engineer

RAGFLOW
BY THE SPECS

A serious system on modest hardware. **Docker compose up** and you're running the full stack.

80.3K	11 / 10	4 / 16
GITHUB STARS	TEMPLATES / LAYOUT TYPES	CORES / GB MINIMUM

BYO LLM: GPT-5, Gemini 3 Pro, DeepSeek v4, Claude, Ollama. **BYO store:** Elasticsearch (default) or Infinity. **Connectors:** Confluence, Notion, S3, Google Drive, Discord — synced on a schedule, no glue code.

PULKIT TYAGI
Senior Gen-AI Engineer

RAGFLOW VS FRAMEWORKS

Different optimization targets. **Pick by where your hard part lives** — document quality, or pipeline flexibility.

RAGFLOW	LANGCHAIN / LLAMAINDEX
Parser-first, opinionated stack	Libraries, assemble components
DeepDoc + TSR ship by default	BYO parser (PyPDF, Unstructured)
Visual chunks, grounded UI citations	Citations & UI are your problem
GraphRAG, agents, MCP built in	Agents via add-ons / LangGraph
Low-code; deploys as one system	Max flexibility, much more code

PULKIT TYAGI
Senior Gen-AI Engineer

SELF-HOST IT IN MINUTES

Clone, configure, compose. **Elasticsearch + MinIO + Redis + MySQL** come bundled — one command boots the lot.

deploy-ragflow.sh

```
# 1. Clone — requires Docker ⩾ 24, Compose ⩾ 2.26
git clone https://github.com/infiniflow/ragflow.git
cd ragflow/docker

# 2. Pick a doc store: Elasticsearch (default) or Infinity
#    Set LLM factory + API key in service_conf.yaml.template
#    (GPT-5, Gemini 3 Pro, DeepSeek v4, Claude, Ollama)

# 3. Boot the full stack — ES, MinIO, Redis, MySQL, RAGFlow
docker compose -f docker-compose.yml up -d

# 4. Open the UI, create a knowledge base, upload a PDF
open http://localhost:80
```

RUN IT ON ONE REAL PDF THIS WEEK.

LIKE COMMENT SHARE SAVE